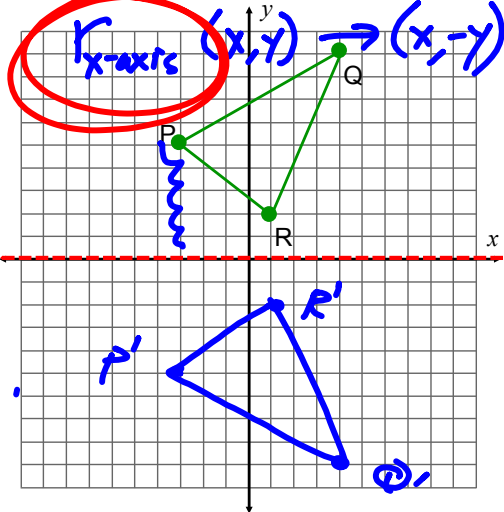


Students will perform rotations about the origin.

Do Now

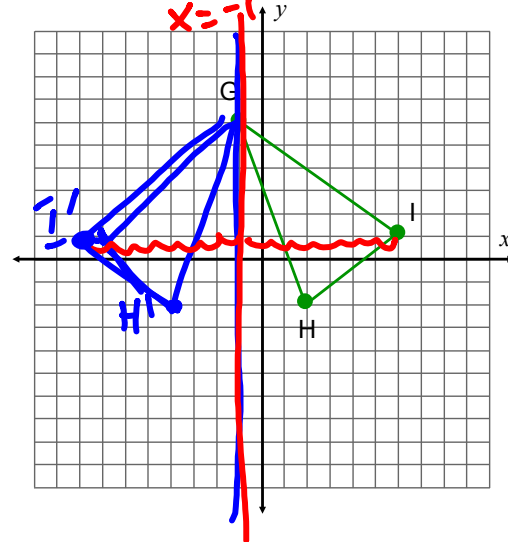
1) State the coordinates of the vertices of $\triangle P'Q'R'$, the image of $\triangle PQR$ after being reflected over the x -axis.

$P(1, -2)$ $Q(4, -4)$ $R(-3, -5)$



2) State the coordinates of the vertices of $\triangle G'H'I'$, the image of $\triangle GHI$ after being reflected over the line $x = -1$.

$G'(-1, 6)$ $H'(-4, -2)$ $I'(-8, 1)$



Rotations

A rotation is a rigid motion in which every point in a figure is moved in a circle around a fixed point, called the center of reflection.

Every rotation requires 3 crucial pieces of information:

1. The center of the rotation.

2. The direction of the rotation.

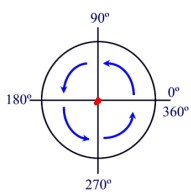
Will always be either CW or CCW

If no direction given by others, assume CCW

3. How far to rotate, given as a degree or a ending point.

Rotations are written as R_d , where d is the number of degrees, usually in multiples of 90: 90° , 180° , 270° , or 360° .

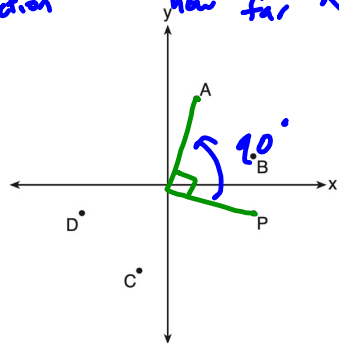
A rotation of 360° will bring a shape back onto its original starting position.



Do Later

5 Which point shown in the graph below is the image of point P after a counterclockwise rotation of 90° about the origin?

direction how far center

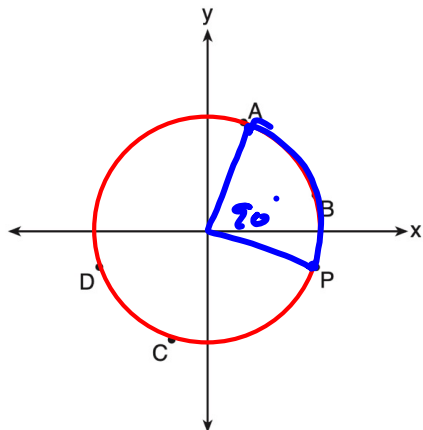


- (1) A
(2) B

- (3) C
(4) D

Do Later

5 Which point shown in the graph below is the image of point P after a counterclockwise rotation of 90° about the origin?

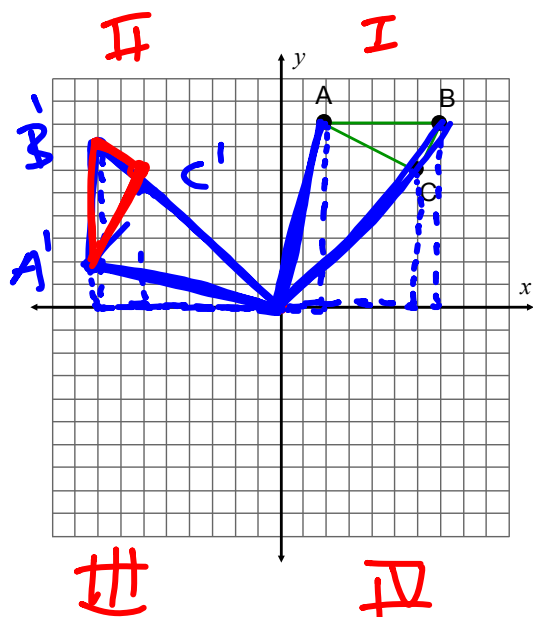


- (1) A
(2) B

- (3) C
(4) D

90° Rotations About Origin

Ex 1: Graph, label, and state the coordinates of $\triangle A'B'C'$, the image of $\triangle ABC$ after being rotated 90° counterclockwise about the origin.



Coordinate rule

$$R_{90^\circ}(x, y) = (-y, x)$$

Tip:

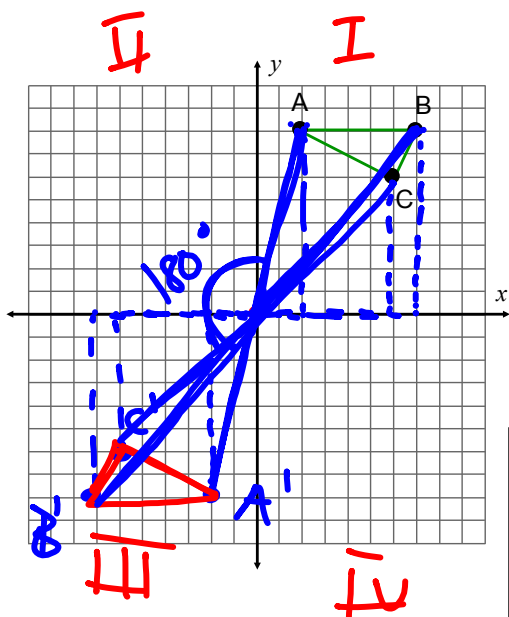
$$A(2, 8) \longrightarrow A'(-8, 2)$$

$$B(7, 8) \longrightarrow B'(-8, 7)$$

$$C(6, 6) \longrightarrow C'(-6, 6)$$

180° Rotations About Origin

Ex 2: Graph, label, and state the coordinates of $\triangle A'B'C'$, the image of $\triangle ABC$ after being rotated 180° counterclockwise about the origin.



Coordinate rule

$$R_{180^\circ}(x, y) = (-x, -y)$$

$$A(2, 8) \longrightarrow A'(-2, -8)$$

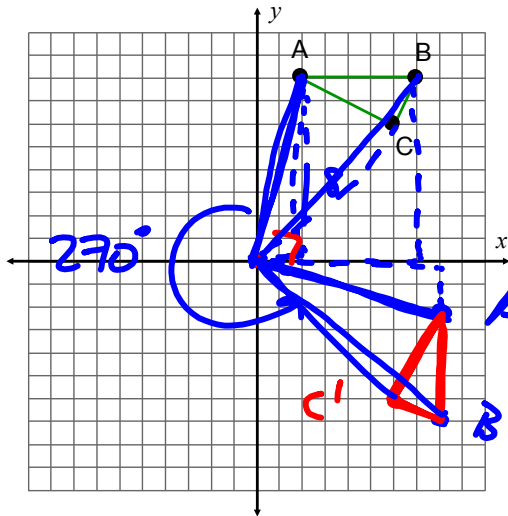
$$B(7, 8) \longrightarrow B'(-7, -8)$$

$$C(6, 6) \longrightarrow C'(-6, -6)$$

Note: A 180° rotation is the same as a reflection through a point

270° Rotations About Origin

Ex 3: Graph, label, and state the coordinates of $\triangle A'B'C'$, the image of $\triangle ABC$ after being rotated 270° counterclockwise about the origin.



Coordinate rule

$$R_{270^\circ}(x, y) = (y, -x)$$

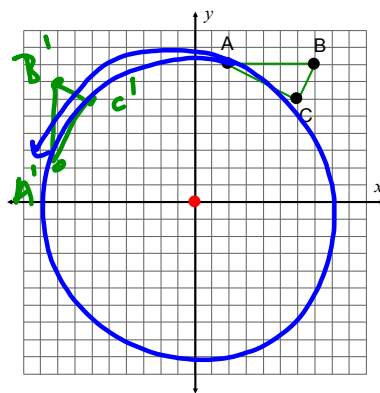
$$A(2, 8) \longrightarrow A'(8, -2)$$

$$B(7, 8) \longrightarrow B'(8, -7)$$

$$C(6, 6) \longrightarrow C'(6, -6)$$

Rotations by Angles $\geq 360^\circ$ About Origin

Ex 4: Graph, label, and state the coordinates of $\triangle A'B'C'$, the image of $\triangle ABC$ after being rotated 450° counterclockwise about the origin.



Rotating a figure more than 360° is the same as rotating it by the angle minus a multiple of 360° .

$$450^\circ - 360 = 90^\circ$$

$$R_{450} = R_{90}$$

$$A(2, 8) \longrightarrow A'(-8, 2)$$

$$B(7, 8) \longrightarrow B'(-8, 7)$$

$$C(6, 6) \longrightarrow C'(-6, 6)$$

Memorizing Coordinate Rules For Rotations

One way to help memorize the coordinate rules for rotations is to remember that to rotate 90° counterclockwise about the origin:

"Change the sign of y and switch positions of coordinates."

$$(x, y) \xrightarrow{R_{90^\circ}} (-y, x) \xrightarrow{R_{180^\circ}} (-x, -y) \xrightarrow{R_{270^\circ}} (y, -x)$$

$$(2, 8) \xrightarrow{R_{90^\circ}} (-8, 2) \xrightarrow{R_{180^\circ}} (-2, -8) \xrightarrow{R_{270^\circ}} (8, -2)$$

$$(-3, 2) \xrightarrow{R_{90^\circ}} (\quad , \quad) \xrightarrow{R_{180^\circ}} (\quad , \quad) \xrightarrow{R_{270^\circ}} (\quad , \quad)$$

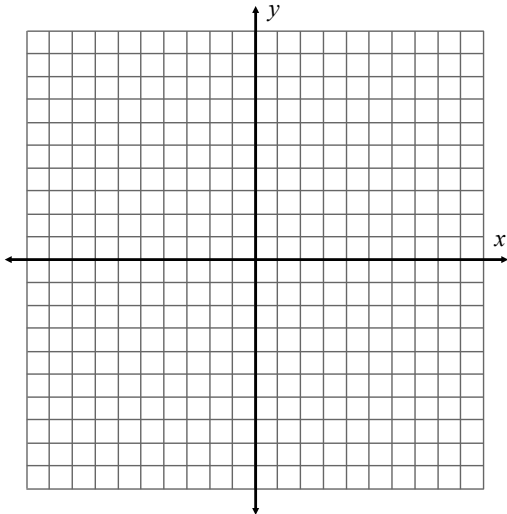
$$(-4, -5) \xrightarrow{R_{90^\circ}} (\quad , \quad) \xrightarrow{R_{180^\circ}} (\quad , \quad) \xrightarrow{R_{270^\circ}} (\quad , \quad)$$

Ex 5: Complete the missing information in each reflection below.

$R_{90^\circ}(x, y) = (-y, x)$	$R_{90^\circ}(4, 3) = (-3, 4)$
$R_{90^\circ}(-5, -2) = (2, 5)$	$R_{90^\circ}(7, -1) = (1, 7)$
$R_{180^\circ}(x, y) = (-x, -y)$	$R_{180^\circ}(4, 3) = (-4, -3)$
$R_{180^\circ}(-5, -2) = (5, 2)$	$R_{180^\circ}(7, -1) = (-7, 1)$
$R_{270^\circ}(x, y) = (y, -x)$	$R_{270^\circ}(4, 3) = (3, -4)$
$R_{270^\circ}(-5, -2) = (-2, 5)$	$R_{270^\circ}(7, -1) = (-1, -7)$
$R_{90^\circ}(-6, 8) = (-8, -6)$	$R_{180^\circ}(-6, 8) = (6, -8)$
$R_{270^\circ}(-6, 8) = (8, 6)$	$R_{360^\circ}(-6, 8) = (-6, 8)$

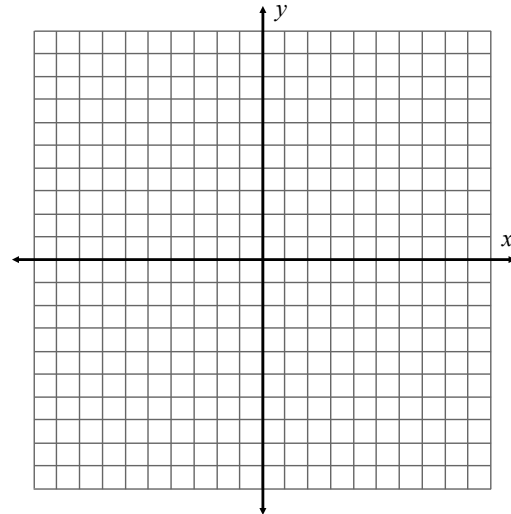
1) The coordinates of the vertices of $\triangle ABC$ are $A(8,1)$, $B(8,-5)$, & $C(-3,-5)$. Graph $\triangle ABC$, then graph, label, and state the coordinates of $\triangle A'B'C'$, the image of $\triangle ABC$ after being rotated 90° CCW about the origin.

$A'(\quad , \quad)$ $B'(\quad , \quad)$ $C'(\quad , \quad)$



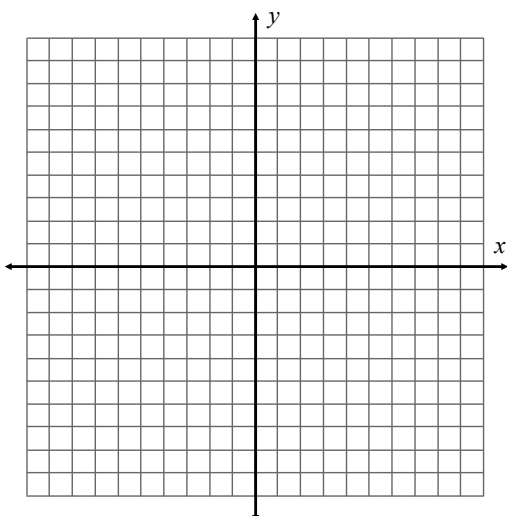
2) The coordinates of the vertices of $\triangle DEF$ are $D(-3,2)$, $E(6,2)$, & $F(0,-4)$. Graph $\triangle DEF$, then graph, label, and state the coordinates of $\triangle D'E'F'$, the image of $\triangle DEF$ after being rotated 180° CCW about the origin.

$D'(\quad , \quad)$ $E'(\quad , \quad)$ $F'(\quad , \quad)$



3) The coordinates of the vertices of $\triangle GHI$ are $G(-1,3)$, $H(7,-1)$, & $I(7,-4)$. Graph $\triangle GHI$, then graph, label, and state the coordinates of $\triangle G'H'I'$, the image of $\triangle GHI$ after being rotated 270° CCW about the origin.

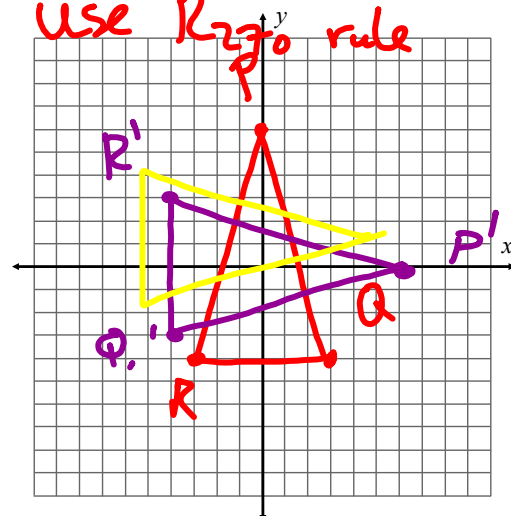
$G'(\quad , \quad)$ $H'(\quad , \quad)$ $I'(\quad , \quad)$



4) The coordinates of the vertices of $\triangle PQR$ are $P(0,6)$, $Q(3,-4)$, & $R(-3,-4)$. Graph $\triangle PQR$, then graph, label, and state the coordinates of $\triangle P'Q'R'$, the image of $\triangle PQR$ after being rotated 90° **clockwise** about the origin.

$P'(6,0)$ $Q'(-4,-3)$ $R'(-4,3)$

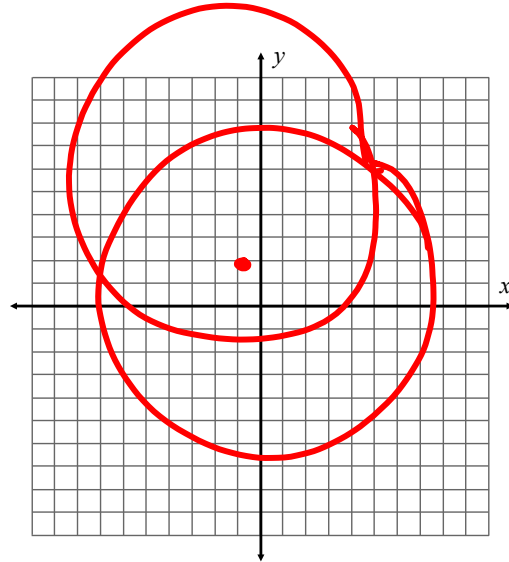
Use R_{270} rule



Challenge

5) The coordinates of the vertices of $\triangle PQR$ are $P(0,6)$, $Q(3,-4)$, & $R(-3,-4)$. Graph, label, and state the coordinates of $\triangle P'Q'R'$, the image of $\triangle PQR$ after being rotated 90° CCW about the point $(-1,2)$.

$P'(\quad, \quad)$ $Q'(\quad, \quad)$ $R'(\quad, \quad)$



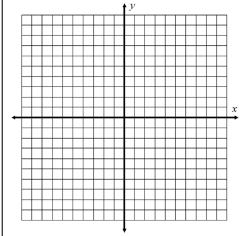
Extension

Segment MN has endpoints $M(-5,3)$ and $N(7,8)$. Find the coordinates of the endpoints of segment $M'N'$, the image of segment MN under the transformation R_{90° .

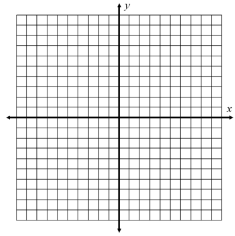
Show that the length of the segment did not change under this transformation.

Assignment 2.01: Reflections

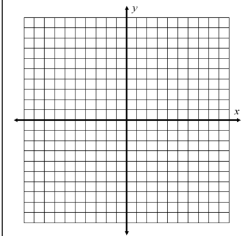
1. Triangle ABC has vertices $A(1,8)$, $B(5,3)$, and $C(-2,1)$. Find the coordinates of the vertices of $\Delta A'B'C'$ after reflection $r_{x\text{-axis}}$.



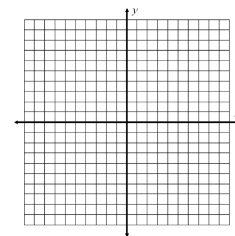
2. Triangle DEF has vertices $D(3,7)$, $E(9,2)$, and $F(6,-1)$. Find the coordinates of the vertices of $\Delta D'E'F'$ after being reflected over the line $y = 2$.



3. Triangle GHI has vertices $G(0,4)$, $H(6,0)$, and $I(2,-5)$. Find the coordinates of the vertices of $\Delta G'H'I'$ after reflection $r_{y\text{-axis}}$.



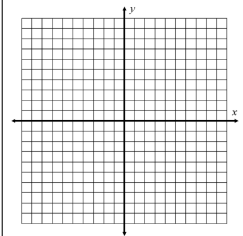
4. Triangle PQR has vertices $P(-6,9)$, $Q(2,5)$, and $R(-7,-1)$. Find the coordinates of the vertices of $\Delta P'Q'R'$ after reflection over the line $x = -3$.



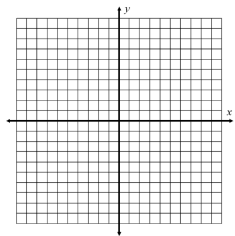
5. Fill in the missing information below:

$r_{x\text{-axis}}(8,-3) = (\quad , \quad)$	$r_{x\text{-axis}}(-2,7) = (\quad , \quad)$
$r_{y\text{-axis}}(8,-3) = (\quad , \quad)$	$r_{y\text{-axis}}(-2,7) = (\quad , \quad)$
$r_{y=x}(8,-3) = (\quad , \quad)$	$r_{y=x}(-2,7) = (\quad , \quad)$
$r_{y=-x}(8,-3) = (\quad , \quad)$	$r_{y=-x}(-2,7) = (\quad , \quad)$

6. Triangle ABC has vertices $A(5,-3)$, $B(9,0)$, and $C(7,5)$. Find the coordinates of the vertices of $\Delta A'B'C'$ after reflection $r_{y=x}$.



7. Triangle DEF has vertices $D(-3,1)$, $E(-5,5)$, and $F(0,0)$. Find the coordinates of the vertices of $\Delta D'E'F'$ after reflection $r_{y=-x}$.



Geometry

Hr:

Name: _____